

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY 2021 TRIMESTER

FINAL ASSESSMENT

UCCC2063 ALGORITHMS ANALYSIS

WEDNESDAY, 15 SEPTEMBER 2021

TIME : 2.00 PM – 5.00 PM (3 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

Instructions to Students:

REMINDER: You are reminded to read and adhere to the Final Assessment Instructions to Candidates that has been made available through the UTAR Portal before the commencement of Final Assessment (FA). The detailed instructions for this FA are as follows:

General

1. This Final Assessment (FA) is an Individual and Time Restricted assessment which consists of FOUR (4) questions. Each question carries 25 marks.
2. You are required to answer **ALL** questions, and submit the **ANSWER SCRIPT** by **5.00 pm, 15 SEPTEMBER 2021**.
3. During the period of **3 hours** of this FA, the examiner(s) can be reached at
 - (a). Microsoft Teams with Code: **q42xncw**, or
 - (b). Email: **tsesh@utar.edu.my**You may use the above e-platform(s) to check with the examiner(s) if you need any clarification on this FA question paper.
4. You may refer to any books, lecture notes, published materials, online resources etc. when answering the questions. Candidates are reminded that **Copy-and-Paste, Consultation, Discussion and Sharing of Answers** with others are **STRICTLY PROHIBITED** in this FA.

UCCC2063 ALGORITHMS ANALYSIS**Answer Script File**

5. The answer script **MUST** be either a Microsoft Word or PDF in A4 size format. **Note: Please keep the file size NOT exceeding 5MB.**
6. Please check your index number generated by the Division of Examinations, Awards, and Scholarships (DEAS). You **MUST** name your answer script file using the following file name for submission:

UCCC2063_FA_[Programme Abbreviation]_[Your Index Number]

For example, if you are from the degree programme CS, and your Index Number is A01234CBCSF, then your answer script should be named as

UCCC2063_FA_CS_A01234CBCSF.doc

Answer Script File Submission

7. Your answer script file has to be submitted to the following platform(s) before the due time/date.
 - (a). Submit your answer script at WBLE.
 - (b). Send your answer script to the course code email:
UCCC2063.FICT@utar.edu.my
Note: For the subject / title of your email, use the file name of your answer script, i.e.,
UCCC2063_FA_CS_[Your Index Number].doc
UCCC2063_FA_CT_[Your Index Number].doc
8. Please make sure you submit the correct, complete and final version of your answer script and the same copy of answer script file is to be submitted to the above platform(s). If different answer scripts are received, the examiner will just randomly choose one of them to mark and the other will be totally ignored.

UCCC2063 ALGORITHMS ANALYSIS

Contents of Answer Script

9. The **first page** of your submission is the **Final Assessment Cover Page**. You **MUST** use the template given and fill in the following information
 - Your Degree Programme (Abbreviation)
 - Your Index Number
 - Your Name
 - Your Student ID
10. The **second page** of your submission is the **Final Assessment Declaration Statement**. You **MUST** use the template given, and digitally sign on the form to indicate the authenticity of your submitted work is without plagiarism.
11. Each question should be answered starting on a new page. It is highly recommended that the answers are placed in same order as the questions.
12. For answer scripts that have text-based answers only, all texts **MUST** be typed and recommended using Times New Roman characters with font size [**e.g. 12**], with proper spacing and alignment, except for the drawings and equations/calculations.
13. For answer scripts that require / contain drawings, equations and calculations with short text descriptions, you can hand-write your answers on a blank paper, and then use the scanner apps in your smartphone (Scanner Document, CamScanner, Adobe Scanner etc.) to take scanned copy, or you can type if necessary and include the scanned copy taken in the Word document, as part of your submitted answers.
14. Please include a page number on **each and every page** of your answer script. Ensure that each page of answer scripts is in sequence prior to online submission.

WARNING OF PLAGIARISM

15. Answer scripts shall be uploaded to Turnitin to check for the originality of submitted answers. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have plagiarized in the answer submitted. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

UCCC2063 ALGORITHMS ANALYSIS**Questions: [Total: 100 Marks]****Q1. [Asymptotic Notations]**

- (a) Express the following functions in terms of Ω notation.
- (i) $3n^{0.5} \log n^{10} + n^{0.9}$ (1 mark)
 - (ii) $(700n)^{763} + n(\log_2 n)^{123} + 0.0001(2^n) + 2^{1.01n}$ (1 mark)
 - (iii) $2(n+1)(n-1)! + 2^n + n^{700} + n!$ (1 mark)
 - (iv) $60n^2 + 939n^{1.8} \log_2 n + 2021$ (1 mark)
 - (v) $\log_{10} n^{2021}$ (1 mark)
 - (vi) $\log_{100} n^{100} + \log^2 n$ (1 mark)
- (b) Using the definition of Big-Oh, prove $2n = O(n^{1.001})$. (6 marks)
[Hint: $O(f(n)) = \{g(n) \mid \exists c, n_0 \bullet \forall n > n_0 \bullet g(n) \leq c f(n)\}$]
- (c) Using the definition of Ω , prove $2n \neq \Omega(n^{1.001})$. (6 marks)
[Hint: $\Omega(f(n)) = \{g(n) \mid \exists c, n_0 \bullet \forall n > n_0 \bullet g(n) \geq c f(n)\}$]
- (d) Prove $\forall c \bullet \exists n_0 \bullet \forall n > n_0 \bullet n^{10000} < c n!$. (7 marks)
[Hint: Note that c is defined by the universal quantifier.]

[Total : 25 marks]

UCCC2063 ALGORITHMS ANALYSIS**Q2. [Solving Recurrence]**

- (a) Consider the recurrence $T(n) = 4T(n/4) + n$.
- (i) Solve $T(n)$ by recursive method (also called expansion). Give your answer in Θ notation. (4 marks)
 - (ii) Solve $T(n)$ by substitution (also called Mathematical Induction). Give your answer in Big-Oh notation. (4 marks)
- (b) Using master method to solve the following recurrences.
- (i) $T(n) = 4T(n/4) + n \log n$. (4 marks)
 - (ii) $T(n) = 4T(n/4) + n^{0.9} \log n$ (4 marks)
 - (iii) $T(n) = 4T(n/4) + n^2 \log n$. (4 marks)
- (c) Let $f(n)$ be an asymptotically positive function. A statement is given below:
- If the logical expression $\exists \varepsilon > 0 \bullet f(n) = O(n^{\log_b a - \varepsilon})$ is true, then the logical expression $\exists \varepsilon > 0 \bullet f(n) = \Omega(n^{\log_b a + \varepsilon})$ must be false.
- (i) Without proving the above statement, what is its implication in solving the recurrence $T(n) = aT(n/b) + f(n)$? (1 mark)
 - (ii) Prove the above statement. (4 marks)
[Hint: No need to use the definitions of Big-Oh and Ω .]

[Total : 25 marks]

UCCC2063 ALGORITHMS ANALYSIS**Q3. [Binary Tree, Fundamentals of Algorithm]**

A leaf node is a node having no children nodes. The recursive definitions of the number of nodes, and the number of leaf nodes, in a binary tree T are as below:

Definition 1: If T is empty, then the number of nodes is zero. If T is not empty, then the number of nodes equals to the sum of the number of nodes on its left and right subtrees, plus one.

Definition 2: If T is empty, then the number of leaf nodes is zero. If $|T| = 1$, then the number of leaf nodes is one. If $|T| > 1$, then the number of leaf nodes equals to the sum of the number of leaf nodes on its left and right subtrees.

- (a) Based on Definition 1, give a recursive algorithm which accepts a binary tree as input, and returns its number of nodes. (3 marks)
- (b) Based on Definition 2, give a recursive algorithm which accepts a binary tree as input, and returns its number of leaf nodes. (4 marks)
- (c) An internal node of a binary tree is a node that has at least one child node. By using the algorithms in (a) and (b), give an algorithm which accepts a binary tree as input, and returns its number of internal nodes. (3 marks)
[Hint: Even if (a) and (b) are not completed, you can use them here.]
- (d) Give a non-empty binary tree such that the number of leaf nodes is less than the number of internal nodes. (2 marks)
- (e) Give a non-empty binary tree such that the number of leaf nodes is equal to the number of internal nodes. (1 mark)
- (f) Give a non-empty binary tree such that the number of leaf nodes is larger than the number of internal nodes. (2 marks)
- (g) A software engineer wants to test for the correctness of the algorithm in (c). He chooses a necessary condition as follows:

Inside any binary tree, the number of leaf nodes is always no more than the number of internal nodes plus one.

Prove the above statement by induction on the number of internal nodes.

(10 marks)

[Hint: Mathematical induction includes basis, inductive, and conclusive steps.]

[Total : 25 marks]

UCCC2063 ALGORITHMS ANALYSIS**Q4. [Design and Analysis of Algorithm]**

- (a) What is the optimal worst-case time complexity for sorting n arbitrarily-ordered elements in array $A[1 .. n]$? (1 mark)
- (b) Name an algorithm that gives the optimal worst case time complexity in (a). (1 mark)
- (c) What is the time complexity of constructing an AVL tree of n elements? (1 mark)
[Hint: AVL tree is a kind of balanced binary search tree.]
- (d) Among preorder, inorder, and postorder traversals on a binary search tree, which one can output the sorted order? (1 mark)
- (e) Among preorder, inorder, and postorder traversals on an AVL tree, which one can output the sorted order? (1 mark)
- (f) You are given an array $A[1 .. n]$ of n integers, and one of them is x . Give an in-place $O(n)$ -time algorithm for partitioning the integers into two groups such that the first group $A[1 .. m]$ contains all non- x integers, and $A[m+1 .. n]$ contains all elements which are equal to x , for some $m \in [1, n-1]$. (5 marks)
[Hint: Get the idea from Hoare's algorithm.]
- (g) Suppose that array $A[1 .. n]$ stores n integers in arbitrary order and there are $n - \left\lceil \frac{n}{\log n} \right\rceil$ identical elements, having value x , inside the array, and the remaining $\left\lceil \frac{n}{\log n} \right\rceil$ elements have arbitrary values.
- (i) Within any $2 * \left\lceil \frac{n}{\log n} \right\rceil + 1$ integers taken from $A[1 .. n]$, what is the minimum number of times that x appears? Explain. (3 marks)
- (ii) Give an $O(n)$ -time algorithm to find out the value of x . Explain why the time complexity is $O(n)$. (5 marks)
[Hint: Apply the ideas of (c), (e), and (g)(i).]

UCCC2063 ALGORITHMS ANALYSIS**Q4. (g) (Continued)**

- (iii) Suppose that $A[1 \dots m]$ contains all the non- x elements, where $m \in [1, n]$. Explain why $O(n)$ -time is enough to sort $A[1 \dots m]$. (3 marks)
[Hint: Just argue for the time complexity, and no need to go into the operations inside any sorting algorithm.]
- (iv) By using the results in (f), (g)(ii), and (g)(iii), give an $O(n)$ -time algorithm for sorting $A[1 \dots n]$. (4 marks)
[Hint: If you cannot finish (f), (g)(ii), or (g)(iii), simply assume you have them done, and use them.]

[Total : 25 marks]